

AI Network Config Diff Reviewer

Enterprise AI-Powered Network Configuration Change Review Platform

Project Overview

AI Network Config Diff Reviewer is an enterprise-grade platform designed to automate network configuration change reviews across multi-cloud environments. The platform compares two versions of network configurations, identifies changes, evaluates security risks, validates compliance requirements, generates AI-powered explanations, and produces approval-ready reports.

Network teams often manage thousands of configuration changes across AWS, GCP, Azure, and Infrastructure-as-Code repositories. Manual review processes are time-consuming, inconsistent, and prone to human error. This solution leverages Artificial Intelligence, rule-based validation, and compliance frameworks to accelerate and standardize network change reviews.

Business Problem

Organizations regularly update:

- AWS Security Groups
- AWS Network ACLs
- AWS Route Tables
- AWS VPN Configurations
- AWS Transit Gateway Routes
- GCP Firewall Rules
- GCP Routes
- GCP VPN Tunnels
- Azure Network Security Groups
- Terraform Infrastructure-as-Code

Manual reviews introduce challenges such as:

- Increased risk of misconfiguration
- Delayed approvals
- Lack of compliance visibility
- Inconsistent risk assessment
- Limited auditability

The AI Network Config Diff Reviewer addresses these challenges through automated analysis and intelligent recommendations.

Project Objectives

The primary objectives of this platform are:

- Detect network configuration changes automatically
 - Identify security risks associated with changes
 - Explain technical modifications in business-friendly language
 - Validate configurations against compliance standards
 - Provide approval recommendations
 - Generate audit-ready reports
 - Maintain a complete review history
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Key Features

Configuration Comparison Engine

The comparison engine analyzes two versions of network configurations and identifies:

Added Configurations

New firewall rules, routes, security group entries, or VPN definitions introduced in the latest version.

Removed Configurations

Network objects or policies removed from the previous version.

Modified Configurations

Changes to:

- Source CIDRs
- Destination CIDRs
- Ports

- Protocols
- Routes
- VPN settings
- Transit Gateway configurations

Supported formats include:

- Terraform (.tf)
 - JSON
 - YAML
 - AWS Export Files
 - GCP Export Files
 - Azure Configuration Exports
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AI-Powered Risk Analysis

The platform integrates with Ollama and Llama 3 to provide intelligent risk assessment.

The AI engine evaluates:

Public Exposure Detection

Example:

Previous CIDR:

10.10.1.0/24

Updated CIDR:

0.0.0.0/0

Risk:

Critical

Reason:

Resource becomes accessible from the public internet.

Sensitive Port Exposure

Detects exposure of critical ports:

- 22 (SSH)
- 3389 (RDP)
- 3306 (MySQL)
- 5432 (PostgreSQL)
- 6379 (Redis)
- 9200 (Elasticsearch)

Route Changes

Detects modifications that impact traffic flow and routing behavior.

VPN Changes

Validates VPN tunnel configuration modifications and encryption settings.

Transit Gateway Changes

Identifies connectivity changes between VPCs and networks.

Compliance Validation Engine

The compliance engine evaluates changes against industry standards.

CIS Controls

Validates against:

CIS AWS Foundations Benchmark

NIST Controls

Validates against:

NIST SP 800-53

PCI-DSS Controls

Validates security controls required for PCI environments.

Organization-Specific Policies

Supports custom enterprise security rules.

Example:

Rule:

"No database port may be exposed to 0.0.0.0/0"

Result:

Violation Detected

Risk Scoring Engine

Each configuration change receives a risk score.

Risk Levels

Low

Minimal impact.

Medium

Requires review.

High

Potential security concern.

Critical

Immediate attention required.

Example:

Security Group Change:

10.10.1.0/24 → 0.0.0.0/0

Risk Score:

Critical

Human Readable AI Explanations

The platform converts technical changes into understandable business language.

Example:

Technical Change:

CIDR expanded from 10.10.1.0/24 to 0.0.0.0/0.

AI Explanation:

Database access has been expanded from an internal application subnet to the public internet. This significantly increases the attack surface and may allow unauthorized access.

Recommendation:

Restrict access to approved application networks only.

Approval Workflow

Supports enterprise change management processes.

Available statuses:

- Draft
- Pending Review
- Approved
- Rejected
- Closed

Reviewers can:

- Approve changes
 - Reject changes
 - Add comments
 - Track review history
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Report Generation

Automatically generates reports in:

PDF

Executive and audit-friendly format.

Markdown

Developer-friendly format.

JSON

Machine-readable format for integrations.

Generated reports include:

- Change Summary
 - Risk Assessment
 - Compliance Findings
 - AI Recommendations
 - Approval Status
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Audit Trail

The system maintains a complete audit history.

Tracked information includes:

- User
 - Timestamp
 - Original Configuration
 - Updated Configuration
 - Risk Analysis
 - Compliance Results
 - Approval Decisions
 - Reviewer Comments
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System Architecture

Workflow:

Network Engineer



Upload Previous Configuration



Upload Updated Configuration



Configuration Parser



Diff Engine



Rule Validation Engine



Risk Scoring Engine



Llama 3 Analysis (Ollama)



Compliance Validation



Approval Workflow



Report Generation



Audit Storage

Technology Stack

Frontend

React

Material UI

TypeScript

React Query

Redux Toolkit

Backend

FastAPI

Python

Pydantic

SQLAlchemy

Artificial Intelligence

Ollama

Llama 3

Custom Prompt Templates

Rule-Based Risk Engine

Database

PostgreSQL

Cache Layer

Redis

Background Processing

Celery

Redis Queue

Authentication

OAuth 2.0

Keycloak

JWT Tokens

Role-Based Access Control

Containerization

Docker

Docker Compose

Orchestration

Kubernetes

Helm Charts

Monitoring

Prometheus

Grafana

OpenTelemetry

Application Pages

Dashboard

Displays:

- Total Reviews
 - Open Reviews
 - High-Risk Findings
 - Compliance Violations
 - Approval Statistics
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Upload Configuration Page

Upload:

- Old Configuration
- New Configuration

Supported formats:

- TF
 - JSON
 - YAML
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Diff Viewer

Displays:

- Added Rules
 - Removed Rules
 - Modified Rules
 - Side-by-Side Comparison
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AI Analysis Page

Displays:

- Summary
 - Security Impact
 - Risk Score
 - Recommendations
 - Approval Recommendation
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Compliance Report Page

Displays:

- CIS Findings
 - NIST Findings
 - PCI Findings
 - Custom Policy Findings
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Approval Workflow Page

Supports:

- Approve
 - Reject
 - Add Comments
 - Escalate Review
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Reports Page

Generate:

- PDF Reports
 - Markdown Reports
 - JSON Reports
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Audit Logs Page

Displays:

- User Activities
- Approval History
- Security Reviews
- Change History

REST APIs

POST /api/upload

Upload configuration files.

POST /api/diff

Generate configuration differences.

POST /api/analyze

Perform AI-based risk analysis.

POST /api/compliance

Execute compliance validation.

POST /api/report

Generate reports.

GET /api/reviews

Retrieve review history.

GET /api/dashboard

Retrieve dashboard metrics.

GET /api/audit

Retrieve audit logs.

Future Enhancements

- Multi-Cloud Risk Correlation
- Azure Support
- ServiceNow Integration

- Jira Integration
 - Slack Notifications
 - Email Notifications
 - GitHub Pull Request Analysis
 - Terraform Plan Review
 - CI/CD Integration
 - RAG-Based Security Knowledge Assistant
 - Security Copilot
 - AI Remediation Suggestions
 - Change Impact Prediction
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Expected Benefits

- 80% Faster Review Process
- Reduced Human Error
- Improved Compliance Visibility
- Consistent Risk Assessment
- Better Audit Readiness
- Enhanced Security Posture
- Faster Change Approval Cycles
- Enterprise-Grade Governance

This platform transforms traditional manual network configuration reviews into an intelligent, scalable, and auditable AI-assisted security review process.